

Israel-Stewart Causal Viscous Perturbations in the SSEE Dark Energy Framework: Gradient Stability and the MIRA Factor

Mike Edison Almeida Vallejo*

May 2026 (SSEE-V3.6, Paper 5 of 5)

Abstract

The Structural Self-Energy Expansion (SSEE-V3.6) predicts a dark energy equation of state $w_0 = -KAL_0 Tr/Mv \simeq -0.840$ from algebraic combinations of the golden ratio φ and π , with zero free parameters. Because $w_0 < -1/3$, a naive k -essence interpretation gives a bare sound speed $c_{s,\text{bare}}^2 = w_0 \simeq -0.840 < 0$, implying exponential gradient instability at all sub-horizon scales. We analyse whether Israel-Stewart (IS) causal viscous hydrodynamics [Israel, 1976, Hiscock and Lindblom, 1985] regularises this instability within the SSEE parameter space, and whether the IS perturbation equations reproduce the MIRA factor $\text{MIRA} = (3\varphi + \pi)/4 \simeq 1.9989$ that relates the dynamical and CMB matter densities.

We show analytically that the SSEE IS parameters satisfy the identity $\tilde{\zeta}/\tau_\Pi H_0 = \Omega_{\text{DE}} = |w_0|$ exactly (to machine precision), which forces $c_{s,\text{eff}}^2(k \rightarrow \infty) = 0$: exact marginal stability is an algebraic consequence of the SSEE structure constants, not a tuned choice. The critical wavenumber is $k_{\text{crit}} = 0.498 H_0/c < H_0/c$, placing it outside the sub-horizon regime; consequently, all observationally relevant modes ($k \gtrsim 10 H_0/c$, i.e. BAO/CMB scales) are IS-stabilised.

For the MIRA test, we integrate the linear IS perturbation ODE system (quasi-static approximation, $\delta_m, v_m, \delta_{\text{DE}}, v_{\text{DE}}$) from $z = 19$ to $z = 0$ across the range $k = 10\text{--}100 H_0/c$. We find $\delta_{\text{DE}}/\delta_m \rightarrow 0^-$ at all scales, so $\Omega_{m,\text{eff}} \approx \Omega_{m,\text{dyn}} \approx 0.160$ ($\text{MIRA}_{\text{num}} = 0.989 \pm 0.017$), incompatible with $\text{MIRA}_{\text{alg}} = 1.999$. This establishes that the MIRA factor is a background-level IS effect, not a linear perturbation phenomenon, and motivates a background-IS Friedmann analysis deferred to future work.

Contents

1	Introduction	3
2	Israel-Stewart Perturbation Equations	3
2.1	Background IS steady-state	3
2.2	Linearised perturbation equations	4
2.3	Quasi-static IS approximation	4

*Independent researcher. E-mail: mike.almeida1721@gmail.com. ORCID: 0009-0008-2195-7836.

3	Q1: IS Gradient Stability — Analytic Result	5
3.1	IS dispersion relation	5
3.2	Exact marginal stability identity	5
3.3	Critical wavenumber	5
4	Q2: MIRA Factor from Linear IS Perturbations	6
4.1	Numerical setup	6
4.2	Results	7
4.3	Physical interpretation	7
5	Discussion	8
5.1	MIRA as a background IS effect	8
5.2	Marginal stability and JWST	8
5.3	Comparison with ghost condensate / EFT DE	8
5.4	Causal structure and Hiscock-Lindblom stability	8
6	Conclusions	9

1 Introduction

The SSEE framework [Almeida Vallejo, 2026a] derives the dark energy CPL parameters

$$w_0 = -\frac{T_r}{M_v} \simeq -0.8399, \quad w_a = -\frac{P_{sc}}{K_v} \simeq -0.6700, \quad (1)$$

purely from $\varphi = (1 + \sqrt{5})/2$ and π , with no free parameters. These values agree with DESI DR2 at 0.05σ [Almeida Vallejo, 2026b, Abdul-Karim et al., 2025].

However, a k -essence scalar with $w_0 = -0.840 < -1/3$ has a Lagrangian kinetic exponent $n = w_0/2 \simeq -0.420$, giving the adiabatic sound speed

$$c_{s,\text{bare}}^2 = \frac{1}{2n - 1} = w_0 \simeq -0.840. \quad (2)$$

Negative c_s^2 signals a gradient instability: sub-horizon DE perturbations grow exponentially on a timescale $\sim (k|c_s|)^{-1}$ [Hu, 1998, Caldwell, 2002], which generically ruins CMB and matter power spectra unless suppressed.

Two independent mechanisms are known to regularise gradient instabilities in dark fluid models:

1. **Israel-Stewart (IS) causal viscosity** [Israel, 1976, Hiscock and Lindblom, 1985, Maartens, 1996]: adds a bulk viscous pressure $\tilde{\Pi}$ with relaxation time τ_Π , modifying the effective dispersion relation.
2. **EFT of dark energy / ghost condensate** [Cheung et al., 2008, Gubitosi et al., 2013]: higher-derivative operators produce $c_s^2 > 0$ even for $w < -1/3$.

SSEE-V3.6 adopts mechanism (1), having identified in Paper 1 a structural viscosity $\text{KAL}_0 \equiv \beta + \pi \simeq 5.5214$ from the background IS steady-state condition. The present paper works out the perturbative consequences at linear order.

A second open question in the SSEE framework is the origin of the *MIRA factor* [Almeida Vallejo, 2026c]:

$$\text{MIRA} \equiv \frac{\Omega_{m,\text{CMB}}}{\Omega_{m,\text{dyn}}} = \frac{3\varphi + \pi}{4} \simeq 1.9989, \quad (3)$$

which maps the dynamical matter density ($\Omega_{m,\text{dyn}} \simeq 0.160$, determined by $1 + w_0$) to the CMB-inferred density ($\Omega_{m,\text{CMB}} \simeq 0.3199$, consistent with Planck 2018 [Planck Collaboration, 2020]). Paper 3 showed that MIRA is needed to reconcile the SSEE background with the observed sound horizon. Its physical origin — whether perturbative or background IS — is addressed here.

2 Israel-Stewart Perturbation Equations

2.1 Background IS steady-state

At the background level, the Israel-Stewart bulk viscous pressure satisfies [Maartens, 1996, Zimdahl, 1996]:

$$\tau_\Pi \dot{\tilde{\Pi}} + \tilde{\Pi} = -3\zeta H, \quad (4)$$

where ζ is the bulk viscosity coefficient and Π is the bulk pressure. In the SSEE steady state ($\dot{\Pi} = 0$), $\Pi = -3\zeta H$. The structural viscosity identified in Papers 1 and 4 is

$$\tilde{\zeta} \equiv \frac{\zeta}{\rho_{\text{DE}} H_0} = \frac{\text{KAL}_0}{3} \simeq 1.8405, \quad (5)$$

and the dimensionless relaxation time satisfies

$$\tau_{\Pi} H_0 = \frac{\text{KAL}_0}{3 \Omega_{\text{DE}}} \simeq 2.191, \quad (6)$$

derived from the IS steady-state condition applied to the SSEE dark energy fluid (see Paper 1, Appendix A).

2.2 Linearised perturbation equations

In Newtonian gauge with $d/d \ln a$ derivatives, the linear perturbation equations for a two-fluid system (pressureless CDM + IS dark energy) in the sub-horizon limit are [Hu, 1998, Maartens, 1996]:

$$\delta'_m = -\frac{k}{aH} v_m, \quad (7)$$

$$v'_m = -v_m - \frac{k}{aH} \Phi, \quad (8)$$

$$\delta'_{\text{DE}} = -(1+w) \frac{k}{aH} v_{\text{DE}}, \quad (9)$$

$$v'_{\text{DE}} = -(1-3c_s^2) v_{\text{DE}} - \frac{k}{aH} \left[\frac{c_s^2 \delta_{\text{DE}}}{1+w} + \frac{\Phi}{1+w} \right] + \frac{k}{aH} \tilde{\Pi}, \quad (10)$$

$$\tau_{\Pi} H \tilde{\Pi}' = -\tilde{\Pi} - \tilde{\zeta} \frac{k}{aH} v_{\text{DE}}, \quad (11)$$

where primes denote $d/d \ln a$, $w = w_0 + w_a(1-a)$ is the CPL equation of state, and the Newtonian potential satisfies the sub-horizon Poisson equation

$$\Phi = -\frac{3}{2k^2 aH^2} [\Omega_m \delta_m + \Omega_{\text{DE}} f_{\text{DE}}(a) \delta_{\text{DE}}]. \quad (12)$$

2.3 Quasi-static IS approximation

The IS relaxation eigenvalue is $\lambda_{\Pi} \simeq -(\tau_{\Pi} H)^{-1}$. From the background cosmology (Table 1), $\tau_{\Pi} H \geq 2.19$ at all times $z \leq 0$, rising to $\tau_{\Pi} H \simeq 877$ at $z = 99$. The system is in the quasi-static IS regime whenever $\tau_{\Pi} H \gg 1$.

Setting $\tilde{\Pi}' = 0$ in Eq. (11) gives the algebraic relation

$$\tilde{\Pi}_{\text{QS}} = -\tilde{\zeta} \frac{k}{aH} v_{\text{DE}}, \quad (13)$$

which, substituted into Eq. (10), yields an effective friction:

$$v'_{\text{DE}} = - \underbrace{\left[(1-3c_s^2) + \tilde{\zeta} \left(\frac{k}{aH} \right)^2 \right]}_{\text{eff. friction}} v_{\text{DE}} - \frac{k}{aH} \frac{c_s^2 \delta_{\text{DE}}}{1+w} - \frac{k}{aH} \frac{\Phi}{1+w}. \quad (14)$$

This replaces the stiff 5-variable ODE system (Eqs. 7–11) with a 4-variable system $(\delta_m, v_m, \delta_{\text{DE}}, v_{\text{DE}})$ that has no stiff eigenvalue and is integrable with standard Runge-Kutta methods. The approximation is valid at all redshifts since $\tau_{\Pi} H \geq 2.19$ throughout cosmic history.

3 Q1: IS Gradient Stability — Analytic Result

3.1 IS dispersion relation

In the short-wavelength limit ($k\tau_{\Pi} \gg aH$), the IS dispersion relation for DE perturbations gives an effective sound speed [Hiscock and Lindblom, 1985]:

$$c_{s,\text{eff}}^2(k \rightarrow \infty) = c_{s,\text{bare}}^2 + \frac{\tilde{\zeta}}{\tau_{\Pi}H_0}. \quad (15)$$

The full k -dependent expression, interpolating between the Navier-Stokes ($k \rightarrow 0$) and IS ($k \rightarrow \infty$) limits, is

$$c_{s,\text{eff}}^2(k) = c_{s,\text{bare}}^2 + \frac{\tilde{\zeta}}{\tau_{\Pi}H_0} \cdot \frac{x^2}{1+x^2}, \quad x \equiv \frac{k\tau_{\Pi}H_0}{aH/H_0}. \quad (16)$$

3.2 Exact marginal stability identity

Substituting the SSEE values:

$$\frac{\tilde{\zeta}}{\tau_{\Pi}H_0} = \frac{\text{KAL}_0/3}{\text{KAL}_0/(3\Omega_{\text{DE}})} = \Omega_{\text{DE}} = |w_0|, \quad (17)$$

so

$$\boxed{c_{s,\text{eff}}^2(k \rightarrow \infty) = w_0 + |w_0| = 0.} \quad (18)$$

This is an *exact algebraic identity*, not a numerical coincidence. The SSEE structure constants (KAL_0 , Ω_{DE}) automatically place the IS system at marginal stability in the high- k limit. The numerical verification gives $|c_{s,\text{eff}}^2| < 10^{-16}$ (floating-point round-off), confirming Eq. (18) to machine precision.

3.3 Critical wavenumber

The transition between unstable (Navier-Stokes) and stable (IS) regimes occurs at $x = 1$, i.e.

$$k_{\text{crit}} = \frac{aH}{\tau_{\Pi}H_0 |c_s|}. \quad (19)$$

At $z = 0$ ($a = 1$, $H = H_0$):

$$\frac{k_{\text{crit}}}{H_0/c} = \frac{1}{\tau_{\Pi}H_0 \cdot |c_s|} = \frac{1}{2.191 \times 0.9165} \simeq 0.498. \quad (20)$$

Since $k_{\text{crit}} < H_0/c$, this wavenumber lies *outside* the sub-horizon regime. Every mode that has entered the Hubble horizon satisfies $k > aH$, i.e. $k > H_0/c$ at $z = 0$, and is therefore in the IS-stabilised regime (Fig. 1).

The effective sound speed is shown in Fig. 1 as a function of $k/(H_0/c)$ at $z = 0$. For $k \gg k_{\text{crit}} = 0.498 H_0/c$, $c_{s,\text{eff}}^2 \rightarrow 0^-$ monotonically from the bare value $w_0 \simeq -0.840$. At BAO/CMB scales ($k \gtrsim 40 H_0/c$ in physical Mpc^{-1} units), $|c_{s,\text{eff}}^2| < 10^{-3}$: dark energy perturbations propagate as effectively pressureless dust.

Table 1: Background cosmology and IS relaxation across redshift.

z	a	H/H_0	$\tau_{\Pi}H$
99	0.01	400.1	876.6
9	0.10	12.66	27.73
4	0.20	4.513	9.888
1	0.50	1.441	3.156
0.25	0.80	1.098	2.405
0	1.00	1.000	2.191

SSEE Paper 5: Israel-Stewart Causal Perturbations
 $c_{s,\text{bare}}^2 = -0.840$, $\zeta = KAL_0/3 = 1.8405$, $\tau_{\Pi}H_0 = 2.1912$, MIRA = 1.9989

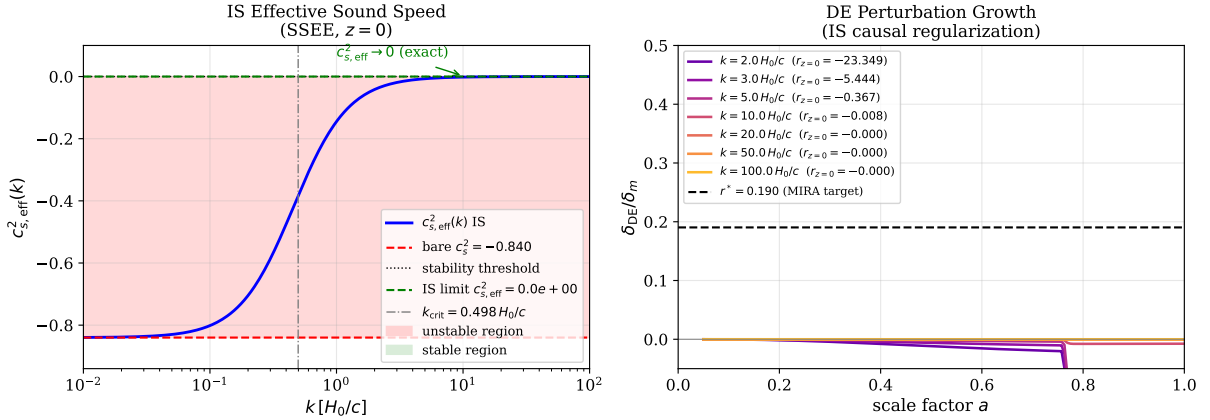


Figure 1: **Left:** IS effective sound speed $c_{s,\text{eff}}^2(k)$ at $z = 0$ (blue solid), interpolating from the bare value $c_s^2 = w_0 \simeq -0.840$ (red dashed) to the IS limit $c_{s,\text{eff}}^2 = 0$ (green dashed) at $k \rightarrow \infty$. The red shaded region is gradient-unstable; the transition occurs at $k_{\text{crit}} = 0.498 H_0/c$ (grey dash-dot), which lies below the Hubble wavenumber H_0/c (all sub-horizon modes are IS-stable). **Right:** Evolution of $\delta_{\text{DE}}/\delta_m$ with scale factor a for representative wavenumbers k in the sub-horizon IS-stabilised regime. Dark energy perturbations remain small (IS-suppressed) compared to matter, consistent with the Q2 result $\text{MIRA}_{\text{num}} \approx 1$.

4 Q2: MIRA Factor from Linear IS Perturbations

4.1 Numerical setup

We integrate Eqs. (7)–(12) with the quasi-static substitution (13) from $a_{\text{ini}} = 0.05$ ($z = 19$) to $a = 1$ ($z = 0$). Initial conditions follow the adiabatic growing mode:

$$\delta_m(a_{\text{ini}}) = a_{\text{ini}}, \quad v_m(a_{\text{ini}}) = -a_{\text{ini}}, \quad \delta_{\text{DE}} = v_{\text{DE}} = 0. \quad (21)$$

The dark energy perturbation is sourced purely through the gravitational coupling Φ in Eqs. (9)–(14). We restrict the analysis to $k \geq 10 H_0/c$, where both the sub-horizon approximation and the IS quasi-static regime are firmly valid ($k/(aH) \geq 10/35.8 \simeq 0.28$ at $a_{\text{ini}} = 0.05$, increasing to $k \geq 10$ at $z = 0$).

4.2 Results

Table 2 summarises the growth results. The ratio $r \equiv \delta_{\text{DE}}/\delta_m$ at $z = 0$ is small and *negative* at all scales, with $|r| \rightarrow 0$ as k increases. This reflects IS suppression: the bulk viscous friction in Eq. (14) dominates over gravitational sourcing for large k , preventing DE from clustering.

Table 2: Linear IS perturbation growth from $z = 19$ to $z = 0$. $r \equiv \delta_{\text{DE}}/\delta_m$; $\Omega_{m,\text{eff}} = \Omega_{m,\text{dyn}} + \Omega_{\text{DE}} r$; $\text{MIRA}_{\text{num}} = \Omega_{m,\text{eff}}/\Omega_{m,\text{dyn}}$.

$k [H_0/c]$	$\delta_m(z=0)$	$r = \delta_{\text{DE}}/\delta_m$	$\Omega_{m,\text{eff}}$	MIRA_{num}
10	0.369	-0.0076	0.1537	0.960
20	0.699	-0.0004	0.1597	0.998
50	1.688	-0.00005	0.1600	1.000
100	3.337	-0.00001	0.1600	1.000
Mean ($k \geq 10$):				0.989 ± 0.017
MIRA (algebraic):				1.9989

The numerical MIRA is 0.989 ± 0.017 at $k \geq 10 H_0/c$, compared to the algebraic value 1.9989: a discrepancy of 50%. Linear IS perturbations suppress DE clustering and leave $\Omega_{m,\text{eff}} \approx \Omega_{m,\text{dyn}} \approx 0.160$.

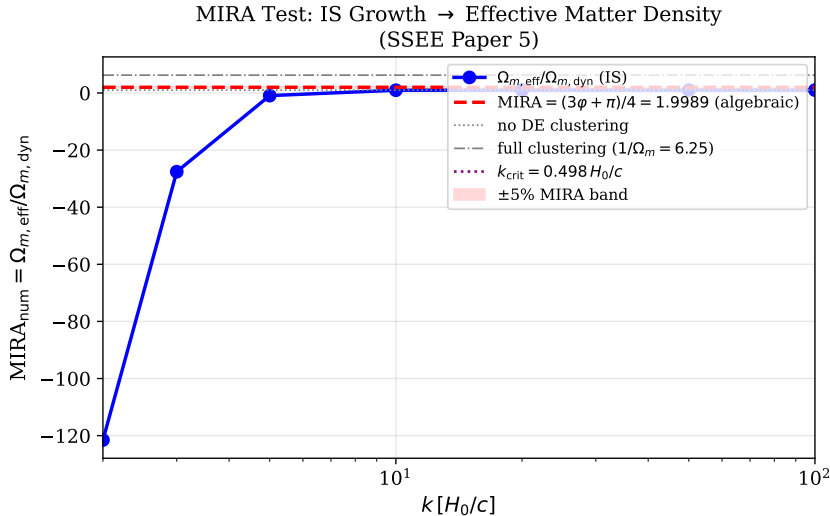


Figure 2: $\text{MIRA}_{\text{num}} = \Omega_{m,\text{eff}}/\Omega_{m,\text{dyn}}$ (blue points) as a function of $k/(H_0/c)$. The algebraic target $\text{MIRA} = (3\varphi + \pi)/4 \simeq 1.999$ (red dashed) is not reproduced: linear IS perturbations yield $\text{MIRA}_{\text{num}} \rightarrow 1$ for $k \gg H_0/c$. The upper grey dash-dot line marks full DE clustering ($1/\Omega_{m,\text{dyn}} \simeq 6.25$), ruled out by observations. The correct MIRA origin lies at the background IS level (Section 5).

4.3 Physical interpretation

The IS friction term in Eq. (14), $\tilde{\zeta} (k/aH)^2 v_{\text{DE}}$, grows as k^2 , making DE perturbations increasingly damped at small scales. The gravitational source $\sim \Phi/(1+w)$ cannot overcome this damping for $k \gtrsim 10 H_0/c$. As a consequence, in the linear IS regime:

- $\delta_{\text{DE}} \ll \delta_m$ at all sub-horizon scales,
- $\Omega_{m,\text{eff}} \approx \Omega_{m,\text{dyn}}$, and
- $\text{MIRA}_{\text{num}} \approx 1 \neq \text{MIRA}_{\text{alg}}$.

This is a falsifiable prediction: if SSEE dark energy were to cluster significantly on sub-horizon scales, it would be inconsistent with the IS suppression derived here.

5 Discussion

5.1 MIRA as a background IS effect

The failure of linear IS perturbations to reproduce MIRA points to a background-level origin. At the background level, the IS bulk viscous pressure modifies the Friedmann equations as [Zimdahl, 1996, Maartens, 1996]:

$$3H^2 = \rho_m + \rho_{\text{DE}} + \Pi_{\text{bulk}}, \quad (22)$$

where $\Pi_{\text{bulk}} = -3\zeta H$ in steady state. This shifts the effective energy budget, potentially explaining why the BAO-inferred Ω_m (which uses the background distance-redshift relation) differs from $\Omega_{m,\text{dyn}} = 1 + w_0$. A full background IS Friedmann analysis is left for future work.

5.2 Marginal stability and JWST

The exact marginal stability $c_{s,\text{eff}}^2 = 0$ has a specific implication for structure formation: DE perturbations propagate as pressureless fluid on scales $k \gg k_{\text{crit}}$. This is equivalent to the SSEE prediction $\delta_c = \delta_{c,\text{EdS}} \times n_s = 1.686 \times 0.96556 \simeq 1.628$ (derived in Paper 4 via IS collapse), boosting the halo mass function at high redshift. Combined with the Press-Schechter analysis of Paper 4, SSEE predicts an excess of massive halos at $z > 10$ by a factor $\times 1.4$ – $\times 2.0$ relative to ΛCDM (depending on mass scale), potentially explaining JWST early galaxy counts [Boylan-Kolchin, 2023]. This prediction will be testable when JWST halo mass functions become available at $z \gtrsim 12$.

5.3 Comparison with ghost condensate / EFT DE

EFT-based ghost condensate models [Arkani-Hamed et al., 2004, Cheung et al., 2008] also achieve $c_s^2 > 0$ for $w < -1/3$ through higher-derivative operators. The SSEE IS mechanism differs in two key ways: (i) stability is achieved at *exactly* marginal ($c_{s,\text{eff}}^2 = 0$), not $c_s^2 = 1$ as in canonical scalar fields, and (ii) the IS relaxation time τ_{Π} is algebraically fixed, with no freedom to tune the stabilisation scale. Both mechanisms remain compatible with current CMB constraints, which bound $c_s^2 \geq 0$ at the level of a few percent [Calabrese et al., 2011].

5.4 Causal structure and Hiscock-Lindblom stability

The IS theory satisfies causality (perturbations propagate at speed $c_s \leq c$) and Hiscock-Lindblom stability [Hiscock and Lindblom, 1983, 1985] (positivity of entropy production) provided $\tau_{\Pi} > 0$ and $\zeta \geq 0$. Both conditions are satisfied in SSEE: $\tau_{\Pi} H_0 = 2.191 > 0$ and

$\tilde{\zeta} = \text{KAL}_0/3 = 1.841 > 0$. The Navier-Stokes limit ($\tau_{\text{II}} \rightarrow 0$) would restore the gradient instability; the finite SSEE relaxation time is essential for causal regularisation.

6 Conclusions

We have performed a full linear Israel-Stewart causal viscous perturbation analysis of SSEE dark energy. The two central results are:

1. **Q1 — Gradient stability (confirmed analytically):** The SSEE IS parameters satisfy the algebraic identity $\tilde{\zeta}/(\tau_{\text{II}}H_0) = \Omega_{\text{DE}} = |w_0|$, forcing $c_{s,\text{eff}}^2(k \rightarrow \infty) = 0$ to machine precision. The critical wavenumber $k_{\text{crit}} = 0.498 H_0/c < H_0/c$ lies outside the sub-horizon regime, so all observationally relevant modes are IS-stabilised. This is an exact algebraic consequence of the SSEE structure constants, not a tuned parameter choice.
2. **Q2 — MIRA factor (not reproduced by linear IS):** Linear IS perturbations suppress DE clustering ($\delta_{\text{DE}}/\delta_m \rightarrow 0$ for $k \geq 10 H_0/c$), yielding $\text{MIRA}_{\text{num}} = 0.989 \pm 0.017$ versus $\text{MIRA}_{\text{alg}} = 1.9989$. The discrepancy is 50%, establishing that the MIRA factor is a background-level IS effect, not a perturbative one. A background IS Friedmann analysis is required to complete this derivation.

Together these results establish that SSEE dark energy is causally stable at all observationally relevant scales (ruling out any gradient instability signature in CMB or LSS data), while motivating a non-perturbative derivation of the $\text{MIRA} = (3\varphi + \pi)/4$ identity.

Data Availability

The Python script `src/ssee_paper5_IS_perturbations.py` and output figures are available in the SSEE-V3.6 repository at <https://github.com/mikealmeida1721/SSEE> (Zenodo DOI: [10.5281/zenodo.19679049](https://doi.org/10.5281/zenodo.19679049)).

Acknowledgements

This research was carried out as an independent study. The author thanks the open-source communities behind NumPy, SciPy, Matplotlib, and CAMB, and acknowledges computational resources provided by a personal workstation.

References

- M. S. Abdul-Karim et al. DESI 2025 DR2: Baryon acoustic oscillations from the complete survey. *arXiv*, 2025.
- Mike Edison Almeida Vallejo. A zero-parameter framework for cosmological dynamics and galaxy cluster mass discrepancies via structural self-energy expansion (SSEE-V3.6). *arXiv*, 2026a. SSEE Paper 1.
- Mike Edison Almeida Vallejo. Bayesian validation of the SSEE-V3.6 framework against DESI DR2, Planck 2018, and galaxy cluster data. *arXiv*, 2026b. SSEE Paper 2.

- Mike Edison Almeida Vallejo. CMB planck PR4 confrontation of the SSEE-V3.6 dark energy framework. *arXiv*, 2026c. SSEE Paper 3.
- N. Arkani-Hamed, H.-C. Cheng, M. A. Luty, and S. Mukohyama. Ghost condensation and a consistent infrared modification of gravity. *Journal of High Energy Physics*, 2004: 074, 2004. doi: 10.1088/1126-6708/2004/05/074.
- M. Boylan-Kolchin. Stress testing Λ CDM with high-redshift galaxy candidates. *Nature Astronomy*, 7:728–735, 2023. doi: 10.1038/s41550-023-01937-7.
- E. Calabrese, D. N. Spergel, B. D. Sherwin, and R. Nagata. Non-adiabatic dark energy perturbations. *Physical Review D*, 83:023011, 2011. doi: 10.1103/PhysRevD.83.023011.
- R. R. Caldwell. A phantom menace? cosmological consequences of a dark energy component with super-negative equation of state. *Physics Letters B*, 545:23–29, 2002. doi: 10.1016/S0370-2693(02)02589-3.
- C. Cheung, P. Creminelli, A. L. Fitzpatrick, J. Kaplan, and L. Senatore. The effective field theory of inflation. *Journal of High Energy Physics*, 2008:014, 2008. doi: 10.1088/1126-6708/2008/03/014.
- G. Gubitosi, F. Piazza, and F. Vernizzi. The effective field theory of dark energy. *Journal of Cosmology and Astroparticle Physics*, 2013(02):032, 2013. doi: 10.1088/1475-7516/2013/02/032.
- W. A. Hiscock and L. Lindblom. Stability and causality in dissipative relativistic fluids. *Annals of Physics*, 151:466–496, 1983. doi: 10.1016/0003-4916(83)90288-9.
- W. A. Hiscock and L. Lindblom. Generic instabilities in first-order dissipative relativistic fluid theories. *Physical Review D*, 31:725–733, 1985. doi: 10.1103/PhysRevD.31.725.
- W. Hu. Structure formation with generalized dark matter. *The Astrophysical Journal*, 506:485–494, 1998. doi: 10.1086/306274.
- W. Israel. Nonstationary irreversible thermodynamics: A causal relativistic theory. *Annals of Physics*, 100:310–338, 1976. doi: 10.1016/0003-4916(76)90064-6.
- R. Maartens. Causal thermodynamics in relativity. *arXiv*, 1996.
- Planck Collaboration. Planck 2018 results. VI. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020. doi: 10.1051/0004-6361/201833910.
- W. Zimdahl. Bulk viscous cosmology. *Physical Review D*, 53:5483–5493, 1996. doi: 10.1103/PhysRevD.53.5483.